

To: Senior Digital Policy Officer, African Union Commission
From: Varun Sen Bahl, Policy Analyst
Date: April 30, 2026
Re: Where to Invest in African NLP: Under-Resourced Languages

A. Executive Summary

Recent investments from the African Union (AU), global philanthropies, AI labs, and others have led to the increased availability of African language datasets for use in artificial intelligence (AI) and natural language processing (NLP). Several researchers and policymakers have noted that these investments have been concentrated in a small number of African languages (such as Swahili, Amharic, Yoruba, Hausa, and Hassaniya Arabic).

Drawing on our analysis of over 2,500 datasets available on the Hugging Face platform, identified by querying languages self-reported by Afrobarometer survey respondents, this memo affirms that and identifies key languages meriting the next round of investment. We call for investment in Southeastern African languages, such as Sudanese Arabic, Borana in Kenya, Anang in Nigeria, Beja in Sudan, as these appear to be the languages with the largest gap between the number of speakers and number of datasets. The AU should leverage its role as a continental-level multilateral organization to direct investments and efforts into these languages.

In addition, existing datasets are concentrated in a narrow set of capabilities—primarily text generation and translation—while critical areas such as speech, question answering, and information retrieval remain underdeveloped. We also identify challenges in the underlying data infrastructure, including inconsistencies in language tagging standards, gaps in ISO-based classification systems, and the mapping of tasks on the Hugging Face platform.

B. Many African Languages Remain Unavailable for NLP & AI

The African continent is home to over 2000 languages (Sands & Gunnink, 2019). The vast majority of these languages have not been recorded in well-structured formats. As a result, they remain underrepresented in the fields of natural language processing (NLP), language modelling, and artificial intelligence (AI) development (Adebara, 2025).

This has practical consequences: generative AI tools (such as chatbots) cannot operate natively in African languages. As African governments and companies increasingly use such tools to deliver information and services across Africa, closing the under-representation of African languages is key to ensuring the accuracy, usefulness, reliability, and inclusivity of these systems. This is well-recognized: the African Union (AU) considers the digitalization of local and indigenous African languages as a top priority (African Union, 2024).

In recent years, a growing number of African language datasets have been developed by African and global researchers, startups, and community organizations, supported by investments from the AU, philanthropies, technology firms, and multilateral initiatives like the SMART Africa Alliance

(Hussen et al., 2026). Projects like the Masakhane Languages Hub, Mozilla's Common Voice, and Africa Next Voices Project are notable examples of successful initiatives (Adebara, 2025)

Yet, despite these gains, progress has been uneven. Researchers examining the NLP research landscape have noted how dataset creation remains concentrated in a relatively small set of widely spoken African languages, including Swahili, Yoruba, and Amharic. Many other languages, that are often not as widely spoken across the continent *but* are used widely within specific countries, continue to be under-represented as data. Recent advances have not yet translated into broad-based linguistic inclusion across the continent.

There is a need to identify which languages need to be prioritized for the next round of investments and partnerships. To identify these languages, this memo turns to an analysis of available language datasets on the popular *Hugging Face Datasets* platform.

C. Hugging Face Datasets as a Source for African Language Datasets

The *Hugging Face Datasets* platform is one of the world's largest repositories of language datasets and AI models (Lhoest et al., 2021). It hosts a rapidly growing collection of over 980,000 publicly available datasets in thousands of languages contributed by developers, organizations, and institutions worldwide, including those from Africa. An analysis of the datasets available on the platform provides a reasonable approximation of the current landscape of data availability in African languages.

While focusing on only this platform may not capture all possible available language datasets, it does provide a consistent basis for assessing availability overall, especially to identify which African languages are available in datasets that can be easily accessed by researchers and developers for NLP and AI research, applications, and other purposes.

The platform's practices make this analysis possible in two ways. First, it permits dataset uploaders to tag datasets using the ISO-639, which is a global standard nomenclature used for language identification and classification (International Organization for Standardization, n.d.). Second, it also has guidelines in place intended to ensure standardized documentation and tagging across datasets. These features enable the systematic identification and comparison of datasets by language availability.

However, our analysis is complicated by the phenomena of multilingualism in Africa; Africans use many different languages across contexts, often combining native tongues or indigenous languages with other more formal or commerce-oriented languages or variants (Adebara, 2025). Africa's colonial legacy and religious landscape means that Africans also often use various globally well-resourced languages like English, French, Portuguese or Arabic. Deciding the range of languages to search for on Hugging Face becomes a key methodological choice that shapes the scope of the analysis.

We resolved this concern by relying on the latest round of the Afrobarometer survey which captures public opinion from a representative sample of citizens from 39 African countries (Felton,

2024; World Bank. n.d.).¹ Survey respondents are asked what they speak at home, giving a list of languages to prioritize. In the latest edition (from 2022), survey respondents reported 657 unique languages, a useful (even if incomplete) set to examine, as these are confirmed to be languages in use. We applied survey weights to approximate the share of a language within a country, and also used, as a secondary measure, global language speaker estimates from the *Joshua Project*, an aggregator of ethnolinguistic statistics (Steuber, 2026).

A second concern also had to be resolved: it is not possible to assign ISO-639 codes to many African languages, as survey respondents often use local names for these languages, names vary significantly across data sources, and language names do not get translated to English consistently. Using manual and automated methods, including a large language model, we were able to map ISO 693 codes to 491 languages in the Afrobarometer survey results. We identified 2512 unique datasets tagged with these codes. Further, several datasets were tagged with multiple languages; these were deliberately counted individually for each language tag, given our focus on identifying under-resourced languages, not just dataset availability in aggregate.

Our analysis reveals a markedly uneven distribution of datasets across languages, reinforcing the under-resourcing noted above: we were also able to identify a set of high-use but under-resourced languages that merit targeted investment. Our key findings are discussed in the next section. [Appendix A](#) details the methodology used to generate the findings we discuss below, while [Appendix B](#) provides a more detailed overview of our findings.

D. Findings: 10% of African Languages have <5 datasets on Hugging Face

Our analysis reveals a markedly uneven distribution of datasets tagged with African languages on Hugging Face. As summary statistics indicate, a handful of languages account for a large share of available datasets: while we identified over 2500 datasets across our sample, the median number of datasets per language is only 15. The most tagged languages were Hassaniya Arabic, Swahili, Amharic, Yoruba, and Hausa (each appearing in 500+ datasets). This is also evident from the scatterplot in Figure 1 below, which plots the number of datasets available for each country–language pair against the global number of speakers (log scale).

¹ These are: Angola, Benin, Botswana, Burkina Faso, Cabo Verde, Cameroon, CongoBrazzaville, Côte d'Ivoire, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guinea, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mauritius, Morocco, Mozambique, Namibia, Niger, Nigeria, São Tomé and Príncipe, Senegal, Seychelles, Sierra Leone, South Africa, Sudan, Tanzania, Togo, Tunisia, Uganda, Zambia, and Zimbabwe (Felton, 2024).

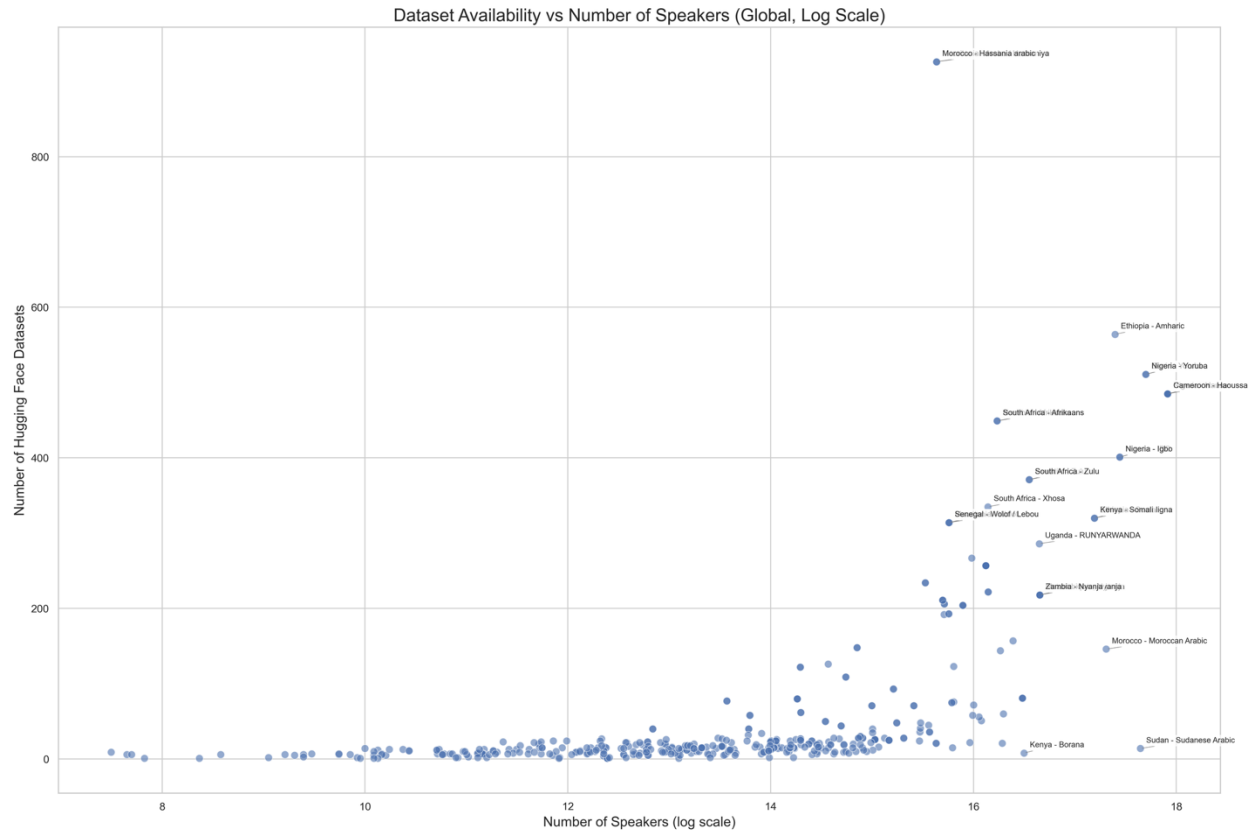


Figure 1: Dataset Availability vs. Number of Speakers (Global, Log Scale)

In terms of under-resourcing, we noted that ~2.4% of languages in scope were not tagged on any dataset at all, while ~10.8% of languages in scope were tagged on fewer than 5 datasets. We first focused on those languages with no datasets tagged that are used by (at least) 1% of survey respondents within their country with no datasets tagged. Table 1 captures the top five:

Rank	Country	Reported Language	Share (%)
1	Malawi	Malawi - Chichewa	71.64
2	Cameroon	Cameroon - Ffululé	12.44
3	Kenya	Kenya - Luhya	11.50
4	Uganda	Uganda - Runyankore	9.39
5	Mozambique	Mozambique - Chichewa	5.92

Figure 2: Top 5 languages with no datasets tagged (per in-country share (%) of use)

We next identified languages with the biggest gap in terms of their in-country usage and the number of datasets tagged with them. A clear outlier is Sudanese Arabic, used by 98% of the Sudanese, but with just 14 tagged datasets (Figures 3 and 4 below). In general, we note the need for investment in Southeastern African languages as listed below.

Rank	Country	Reported Language	Share (%)	Datasets	Gap
1	Sudan	Sudan - Sudanese Arabic	98.10	14	96.59
2	Cabo Verde	Cabo Verde - Creole	99.58	77	91.27
3	Tunisia	Tunisia - Tunisian Arabic	93.36	60	86.88
4	Seychelles	Seychelles - Creole	93.21	77	84.90
5	Lesotho	Lesotho - Sesotho	99.60	206	77.36
6	Eswatini	Eswatini - siSwati	91.95	148	75.97
7	Malawi	Malawi - Chichewa	71.64	0	71.64
8	Morocco	Morocco - Moroccan Arabic	76.85	146	61.09
9	Botswana	Botswana - Setswana	81.28	211	58.49
10	Madagascar	Madagascar - Malagasy avec spécificité régionale	56.57	70	49.02

Figure 3: Top 10 languages with largest gap between datasets and in-country shares of speakers

Rank	Country	Reported Language	Share (%)	Global Speakers	Datasets
1	Sudan	Sudan - Sudanese Arabic	98.10	46,019,600	14
2	Kenya	Kenya - Borana	1.40	14,614,000	8
3	Nigeria	Nigeria - Anang	0.11	3,294,000	12
4	Sudan	Sudan - Beja language	0.87	3,098,000	10
5	Tanzania	Tanzania - Kigogo	1.81	3,008,000	12
6	Nigeria	Nigeria - IZON	0.05	2,783,000	10
7	Nigeria	Nigeria - Ebira	0.04	2,600,000	8
8	Mali	Mali - Malinké	5.57	2,513,800	10
9	Ethiopia	Ethiopia - Hadiyagna	2.07	2,469,000	9
10	Tanzania	Tanzania - Kiha	0.42	2,262,000	9

Figure 4: top 10 languages with the largest gap between global speakers and datasets

The analysis also highlights an uneven distribution in the types of AI capabilities supported. AI capability coverage is uneven: about 18% of datasets support text generation and translation. This is likely because several multilingual datasets were released for these two purposes and were tagged multiple times in the overall dataset. Only ~10% of datasets were tagged with other capabilities (such as named entity recognition, retrieval, speech and question answering). This suggests the importance of investing in specific capabilities as well as languages.

A particularly consequential finding, however, was in the weakness of the ISO-639-3 tags. We were unable to match 193 languages to clear ISO-639 codes. In several cases, a language would be tagged under either (or both) a two-letter ISO code (the ISO-639-1 code used for more commonly used languages) or a three-letter ISO code (the ISO-639-3 code used to comprehensively cover over 7000 languages), as the Hugging Face platform employs both, but not consistently. This made assigning datasets to languages based on their codes inconsistent.

For instance, while our analysis showed ‘no tags’ for the Chichewa language, a manual search on Hugging Face revealed ~120 datasets where this word did appear in the dataset description (however, only ~10 appeared to be clearly language datasets). Many African languages are local variants of larger ‘parental’ groups (e.g., Swahili or Hassaniya Arabic have country-level variants), which are also not tagged consistently. This imprecision is a well-known problem; there have been calls to combine ISO-639-3 codes with country-level codes to improve precision, preserve variation, and tag regional dialects better (Vogt, 2022).

E. Recommendations: Strengthening the African Language Data Landscape

These findings point to clear priorities for targeted investment. To this end, we make the following recommendations for the consideration of your office:

- 1. Invest in Southeastern African languages** as these appear to be the languages with the largest gap between the number of speakers and number of datasets. The African Union should leverage its role as a continental-level multilateral organization to direct investments and efforts into these languages.
- 2. Invest in expanding beyond translation and text-generation datasets**, as current datasets leave African researchers and developers without diverse and sufficient resources between text generation and translation, and especially for speech- or audio-related tasks. As Alabi (2025) notes, AI systems require speech-based capabilities to better serve low-literacy populations across Africa. We affirm this conclusion, as only 9% of datasets were tagged as ‘speech’ capable.
- 3. Invest in improving the standardization of language codes and tags**, especially in terms of enhancing the mapping of ISO-639-1 and ISO-639-3 codes against different names for different languages. The African Union should work with local African developers and NLP experts to create a more granular mapping of codes that recognize country and regional variants of language names (especially as used by local speakers). We also recommend the African Union to work with Hugging Face to improve the tagging of tasks on datasets; ~19% of datasets had no tags, which made analysis challenging.

Appendix A: Methodology

Note: this Appendix is accompanied with Jupyter Notebooks as Annexures, which are available in a Github repository available [here](#). That also contains the datasets used (as described below).

1. Research Question

This analysis seeks to identify which African languages remain under-resourced in the current AI and natural language processing (NLP) ecosystem. Specifically, we ask: *How is the availability of language datasets distributed across African languages, and which widely used languages lack sufficient dataset coverage to support AI applications?* By combining language usage data with dataset availability data on the Hugging Face platform to assess gaps in coverage to answer this question, we aim to inform more targeted and efficient investment in African language data.

2. Methodology

First, we construct a list of languages actively used across African countries using the responses of survey respondents on the latest round (2023) of the Afrobarometer survey, which asks survey respondents, “*what language they speak at home*” enabling us to prioritize specific languages in Africa’s multilingual society. It constructs “*national probability samples designed to generate a sample that is a representative cross-section of all citizens of voting age in a given country_*” (Felton, 2024). Their dataset also gives us a weight (“Combinwt”) that permits the consistent cross-national comparisons of statistics in the dataset, adjusting for the difference in sample sizes across countries (Felton, 2024). We apply this to estimate the in-country share (in %) of a given language as spoken in the location that the survey respondent represents. We then extract a final list of country-language pairs from the survey responses.

Second, we then map the ISO-639-3 and ISO-639-1 codes to the country-language pairs. ISO-639-1 is a standard that assigns two-letter codes to a limited set of widely used languages, such as *en* for English. ISO-639-3 uses three-letter codes to identify thousands of languages globally, including many regional and low-resource languages not covered by ISO-639-1. Notably, these are also how African languages are assigned unique identifiers on the Hugging Face database; their list of the ISO codes used is available on a Github repository [here](#). However, this is not easily mapped to the Afrobarometer dataset, as the name of a language as in the Hugging Face lists does not map in most cases to a survey respondent’s self-identified language.

To address these, we conducted three steps: (1) a cross-referencing of the Hugging Face ISO lists with the Afrobarometer survey data based on the language names (2) a manual mapping of over a 100 ISO codes to language names using a rapid internet search and (3) a third pass using a large language model (GPT 5.2) asking it to map based on its training data and considering only entries where it marked the mapping as “high” level of confidence. To use the LLM, we used Duke’s AI Gateway, which gave an API key to use the above model. Through this process, we successfully map 491 languages to ISO codes. However, 193 languages could not be reliably matched, reflecting broader limitations in existing language classification systems and the representation of African languages within them.

Third, to then assess dataset availability, we count the number of datasets tagged with each language. Where datasets are tagged with multiple languages, they are counted once for each language tag, as the objective is to assess language-level coverage rather than total dataset volume. We then compare language use and dataset availability to identify under-resourced languages, including those with no datasets and those with significant gaps relative to their usage.

Fourth, to assess not just the quantity but also the functionality of available datasets, we categorize datasets based on the AI tasks they support. Using dataset metadata, we group task tags into seven broad categories: generative (e.g., text generation, summarization), translation, classification, labelling (e.g., named entity recognition), retrieval, question answering, and speech. Each dataset is assigned to one or more task categories based on its tags. Where datasets do not match any of the predefined categories, they are classified as “other” or “unknown.” This grouping allows us to evaluate the diversity of capabilities available for each language, in addition to overall dataset counts.

3. Data Sources

We used these datasets:

Variable	Data Source & Details
Dataset availability, language, and task tags on Hugging Face	We use the Hugging Face Datasets platform as a source of publicly available language datasets.
ISO 639-1 and ISO 639-3 codes used on Hugging Face	We use those used by Hugging Face (as available here).
List of African languages in use	We use the Afrobarometer Round 9 survey, which captures language use from a representative sample of respondents across 39 African countries (Felton, 2024).
Global Speaker Estimates	We use global speaker estimates from the Joshua Project as a secondary measure (Steuber, 2026). This dataset only gives the "global speaker count" for a given language, not the per-country speaker count. This is especially challenging if we try to estimate the number of speakers in an African country of a language that is spoken across the world (i.e., beyond just Africa, like French

or Arabic) and even across Africa (e.g. Swahili is spoken in multiple countries). That said, it is less of a concern if we consider that the majority of languages in the first dataset are primarily spoken in those countries.

Note: Ethnologue data is considered to be the leading global resource (see [here](#)) on the number of speakers of a language, but is paywalled, making this dataset the next best available source of this information.

Appendix B: Findings and Patterns Observed

1. Summary Statistics of the Hugging Face Datasets

Metric	Value	Task	Dataset-language Count	Share (%)
Number of countries in scope	38	generative	6901	18.57
Total languages reported in survey	657	translation	5636	15.17
Total languages matched to ISO code	491	classification	3462	9.32
Languages not matched to ISO code	193	labelling	5707	15.36
Total number of datasets found	2512	retrieval	761	2.05
Mean datasets per language	60	qa	2855	7.68
Median datasets per language	15	speech	3539	9.52
Min datasets per language	0	other	1294	3.48
Max datasets per language	926	unknown	7000	18.84
% languages with 0 datasets	2.44			
% languages with <5 datasets	10.79			

2. Top 15 Under resourced Languages (Per Gap Between Datasets & In-country Language Share)

Rank	Country	Reported Language	Share (%)	Datasets	Gap
1	Sudan	Sudan - Sudanese Arabic	98.10	14	96.59
2	Cabo Verde	Cabo Verde - Creole	99.58	77	91.27
3	Tunisia	Tunisia - Tunisian Arabic	93.36	60	86.88
4	Seychelles	Seychelles - Creole	93.21	77	84.90
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6	Eswatini	Eswatini - siSwati	91.95	148	75.97
7	Malawi	Malawi - Chichewa	71.64	0	71.64
8	Morocco	Morocco - Moroccan Arabic	76.85	146	61.09
9	Botswana	Botswana - Setswana	81.28	211	58.49
10	Madagascar	Madagascar - Malagasy avec spécificité régionale	56.57	70	49.02
11	Namibia	Namibia - Oshiwambo (Oshindonga/Oshikwanyama)	45.22	20	43.06
12	Gambia	Gambia - Mandinka	40.52	19	38.47
13	Zimbabwe	Zimbabwe - Shona	65.37	257	37.62
14	Mali	Mali - Bambara	59.33	204	37.30
15	Ghana	Ghana - Akan	53.71	157	36.76

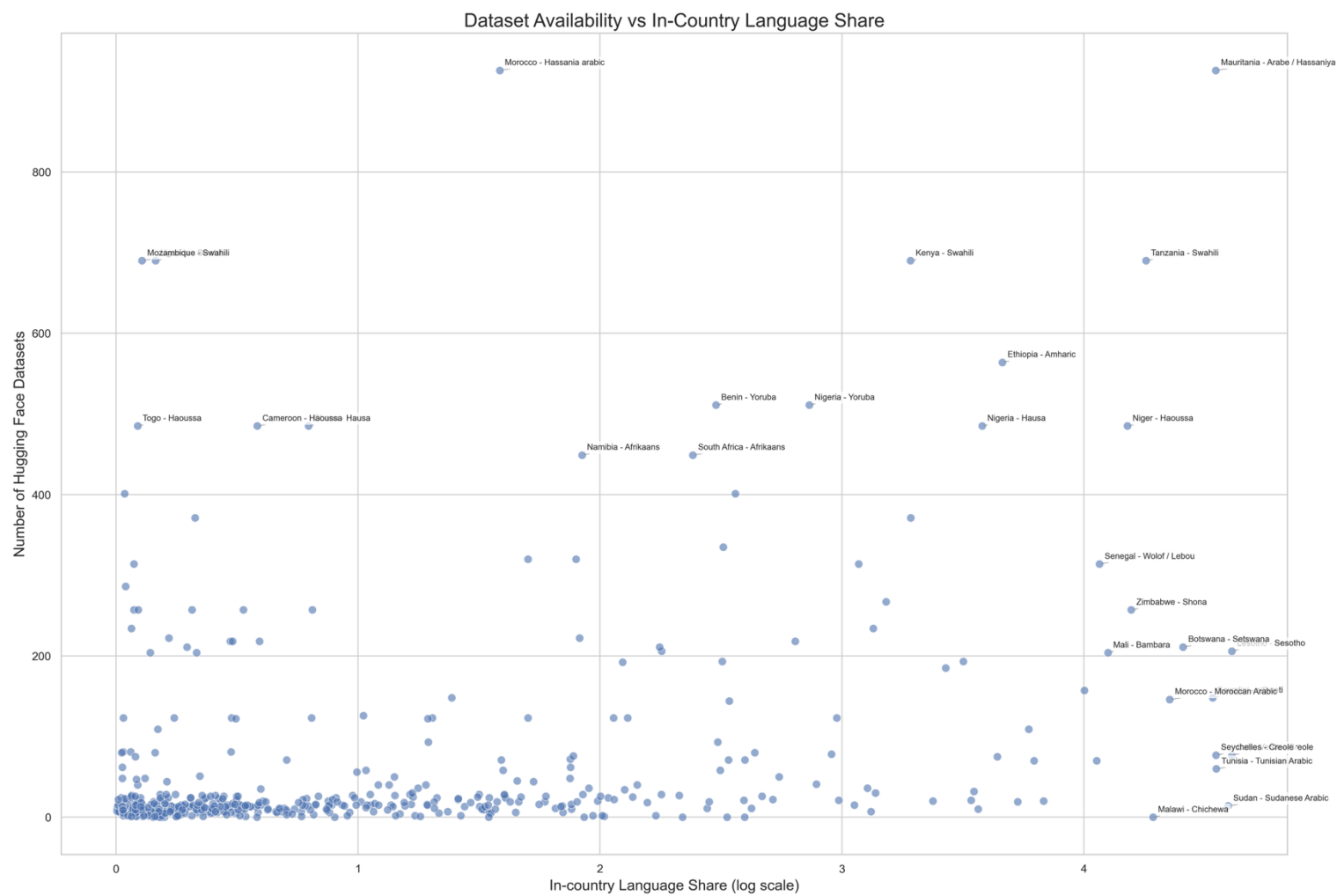
3. Top 15 languages (per in-country share) with no datasets

Rank	Country	Reported Language	Share (%)	Datasets
1	Malawi	Malawi - Chichewa	71.64	0
2	Cameroon	Cameroon - Ffuldéd	12.44	0
3	Kenya	Kenya - Luhya	11.50	0
4	Uganda	Uganda - Runyankore	9.39	0
5	Mozambique	Mozambique - Chichewa	5.92	0
6	Zambia	Zambia - Chewa	3.66	0
7	Cameroon	Cameroon - Bamileke	1.47	0
8	Namibia	Namibia - SAN LANGUAGE	0.79	0
9	Benin	Benin - KOURA	0.22	0
10	Namibia	Namibia - KHWEDAM	0.21	0
11	Sudan	Sudan - Nubian language	0.20	0
12	Guinea	Guinea - Manian	0.17	0

4. List of languages with all capabilities (total of 15)

Rank	Reported Language	Datasets	Capabilities	generative	translation	classification	labelling	retrieval	qa	speech
1	Mauritania - Arabe / Hassaniya	926	7	224	153	162	135	43	158	69
2	Morocco - Hassania arabic	926	7	224	153	162	135	43	158	69
3	Kenya - Swahili	690	7	175	100	110	80	35	116	71
4	Mozambique - Swahili	690	7	175	100	110	80	35	116	71
5	Tanzania - Swahili	690	7	175	100	110	80	35	116	71
6	Uganda - Swahili	690	7	175	100	110	80	35	116	71
7	Ethiopia - Amharic	564	7	133	89	106	88	17	84	74
8	Benin - Yoruba	511	7	115	84	87	75	22	68	57
9	Nigeria - Yoruba	511	7	115	84	87	75	22	68	57
10	Cameroon - Haoussa	485	7	105	97	82	74	16	62	77
11	Ghana - Hausa	485	7	105	97	82	74	16	62	77
12	Niger - Haoussa	485	7	105	97	82	74	16	62	77
13	Nigeria - Hausa	485	7	105	97	82	74	16	62	77
14	Togo - Haoussa	485	7	105	97	82	74	16	62	77
15	Namibia - Afrikaans	449	7	120	91	83	90	28	56	56

5. Plot of Dataset Availability v. In-country Language Share



References

Adebara, I. (2025). *AI and language data flaring in Africa: Addressing the low-resource challenge* (Policy Brief No. 216). Centre for International Governance Innovation.

<https://www.cigionline.org/publications/ai-and-language-data-flaring-in-africa-addressing-the-low-resource-challenge/>

African Union. (2024, July). *Continental artificial intelligence strategy: Harnessing AI for Africa's development and prosperity*. https://au.int/sites/default/files/documents/44004-doc-EN-Continental_AI_Strategy_July_2024.pdf

Alabi, J. O., Hedderich, M. A., Adelani, D. I., & Klakow, D. (2025). Charting the landscape of African NLP: Mapping progress and shaping the road ahead. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* (pp. 27807–27841). Association for Computational Linguistics. <https://aclanthology.org/2025.emnlp-main.1414.pdf>

Felton, J. (2024). *Afrobarometer Round 9 merged data codebook (39 countries, 2021–2023)*. Afrobarometer.

https://www.afrobarometer.org/wp-content/uploads/2024/10/AB_R9.MergeCodebook_25Jun24.final_.pdf

Hussen, K., Sewunetie, W., Ayele, A., Imam, S., Alemu, E., Muhammad, S., & Yimam, S. (2026). *The state of large language models for African languages: Progress and challenges*. In *Proceedings of Machine Learning Research* (Vol. 302, pp. 1–27). <https://proceedings.mlr.press/v302/hussen26a.html>

International Organization for Standardization. (n.d.). *ISO 639 language codes*.

<https://www.iso.org/iso-639-language-code>

Lhoest, Q., Villanova del Moral, A., Jernite, Y., Thakur, A., von Platen, P., Patil, S., Chaumond, J., Drame, M., Plu, J., Tunstall, L., Davison, J., Šaško, M., Chhablani, G., Malik, B., Brandeis, S., Le Scao, T., Sanh, V., Xu, C., Patry, N., McMillan-Major, A., Schmid, P., Gugger, S., Delangue, C., Matussière, T., Debut, L., Bekman, S., Cistac, P., Goehringer, T., Mustar, V., Lagunas, F., Rush, A. M., & Wolf, T. (2021). *Datasets: A community library for natural language processing*. In H. Adel & S. Shi (Eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (pp. 175–184). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2021.emnlp-demo.21>

Sands, B., & Gunnink, H. (2019). Languages of Africa. In *The SAGE encyclopedia of human communication sciences and disorders* (Vol. 4, pp. 1020-1024). SAGE Publications, Inc., <https://doi.org/10.4135/9781483380810.n341>

Steuber, L. (2026). *World languages: 7,130 languages with coordinates and features* [Dataset]. Zenodo. <https://doi.org/10.5281/zenodo.18320704>

Vogt, E. (2022, December 28). *Here's why language coding needs an uplift to ISO 639-3*. World Economic Forum. <https://www.weforum.org/stories/2022/12/heres-why-language-coding-needs-an-uplift/>

World Bank. (n.d.). *Afrobarometer data collection*. <http://microdata.worldbank.org/collections/afrobarometer>
